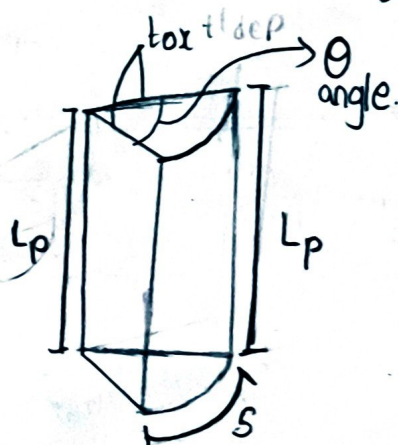
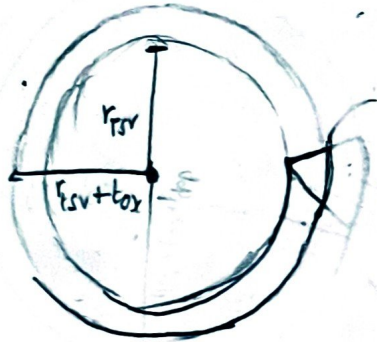
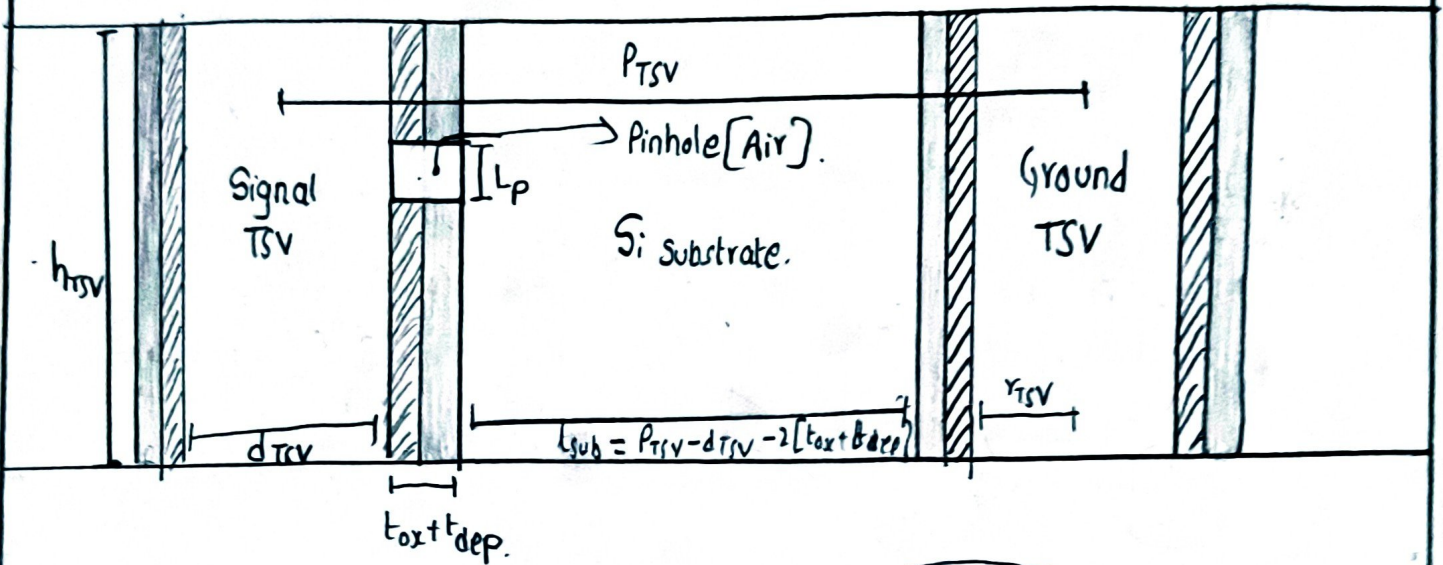
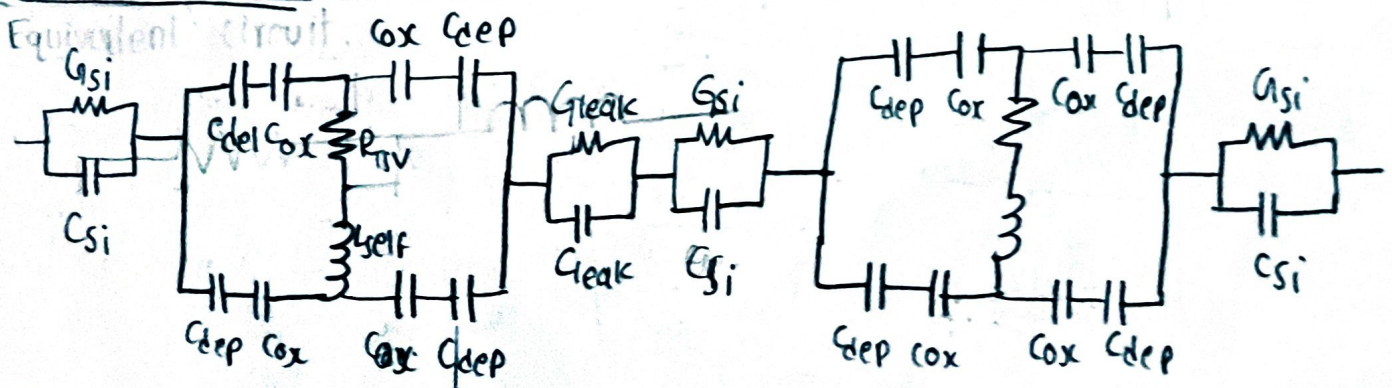


DEFECTED TSV [C_{leak} & G_{leak}]



$s = \text{arclength} \quad \& \quad \theta = \text{radians}$
 $s = r\theta \quad s = [r_{TSV} + t_{ox}] \theta \text{ [m]}$
 $s = [t_{ox} + t_{dep}] \theta \text{ [m]}$

Equivalent circuit.



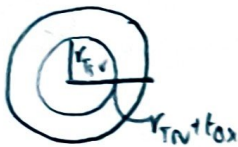
Equations.

$$C_{air} = \frac{2\pi\epsilon L \rho_{air}}{\ln\left[\frac{r_2}{r_1}\right]} \quad \text{if only sector} \quad C_{air} = \frac{\theta}{2\pi} \times \frac{2\pi\epsilon L}{\ln\left[\frac{r_2}{r_1}\right]}$$

where $\ln\left[\frac{r_2}{r_1}\right]$ is C_{ox}

$$C = \frac{\theta \epsilon L}{\ln\left[\frac{r_2}{r_1}\right]}$$

$$C_{air} = \frac{\epsilon_0 \epsilon_{air} \theta L_p}{\ln\left[\frac{r_{TSV} + t_{ox}}{r_{TSV}}\right]}$$



$$C_{leak} = C_{ox} - C_{air}$$

$$= \frac{\pi \epsilon_0 \epsilon_{ox} \left[\frac{h_{TSV}}{2}\right] L_p - \epsilon_0 \epsilon_{air} \theta L_p}{\ln\left[\frac{r_{TSV} + t_{ox}}{r_{TSV}}\right]}$$

$$Y_{leak} = Y_{Si} - Y \text{ [admittive capacitance]}$$

since $G_{air} \cong 0$.

$$Y = j\omega C$$

$$J = \frac{\theta L_p}{2\pi Y_{TSV}}$$

$$G_{leak} = \frac{2\pi \sigma_{Si} L_{TSV} - 4\pi^2 J f \epsilon_0 \epsilon_{air} r_{TSV}}{\ln\left[\frac{r_{TSV} + t_{ox}}{r_{TSV}}\right]}$$

value Table.

Symbol	value
J	10%
θ	$\pi/2$
L_p	12 μm

data taken from a novel TSV model with fault

modelling & simulation of TSV by Chao Liu

Equations

$$R = \frac{\rho l}{A}$$

$$R_{sh} = \frac{\rho_T [L_p + t_{ox}] \left[\frac{\theta}{2} \left[(r_{TSV} + t_{ox})^2 - (r_{TSV})^2 \right] \right]}{[\Omega]}$$

$$\rho_T = \rho_0 [1 + \alpha(T - T_0)]$$

$\theta = \text{radians}$

$$L \approx L_{sh}$$

$$L = \frac{\mu_0 L_p}{2\pi} \left[\ln \left[\frac{2L_p}{r_{eq}} \right] - 1 \right]$$

$$r_{eq} = \sqrt{\frac{A}{\pi}}$$

$$A = \frac{\theta}{2} \left[(r_{TSV} + t_{ox})^2 - (r_{TSV})^2 \right]$$

where

$$A_{sector} = \frac{\theta}{2\pi} (\pi r^2) = \frac{\theta}{2} r^2$$

annular sector

$$A = \frac{\theta}{2} r_2^2 - \frac{\theta}{2} r_1^2 = \frac{\theta}{2} (r_2^2 - r_1^2)$$

Symbol	value
T	298K
T ₀	273K
α	0.039/°C
ρ_0	$1.68 \times 10^{-8} \Omega m$
ρ_T	$3.318 \times 10^{-8} \Omega m$
L _p	12 μm
θ	$\pi/2$

data is taken from paper
 modelling & simulation
 of TSV by chao liu
 A novel TSV model
 by Prosen kiltonia